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1. Which of the following correctly describes the principle of conservation of momentum?
   A. Momentum is the product of mass and velocity.
   B. Momentum is conserved only in elastic collisions.
   C. Momentum is conserved by all bodies in a collision.
   D. Momentum is conserved providing no external force act.

2. A particle of mass m and velocity u is made to collide head-on with a stationary particle of mass M. Given that the collision is elastic, what is the velocity of the particle of mass M after the collision?
   A. u
   B. mu/M
   C. $\dfrac{2mu}{M+m}$
   D. $\dfrac{(M-m)u}{M+m}$

3. In a collision between two steel balls suspended on strings, what are quantities that must be conserved?
   A. kinetic energy, total energy and momentum
   B. kinetic energy and momentum only
   C. kinetic energy and total energy only
   D. total energy and momentum only

> **Solution:** Answer is D
> Reason: total energy and momentum are always conserved. Kinetic energy is conserved only in elastic collision.

4. Two spheres, each of mass m, move with speed v towards each other and have a head-on elastic collision. Which one of the following statements is correct?
   A. The spheres stick together on impact.
   B. The total kinetic energy after impact is mv^2.
   C. The total kinetic energy before impact is zero.
   D. The total momentum before impact is $2mv$

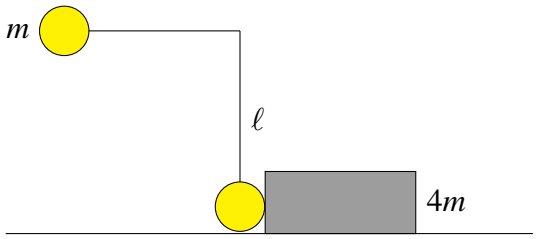
5. A body, initially at rest, explodes and breaks into two masses. The masses, M_1 and M_2, move apart with speeds v_1 and v_2 respectively. What is the ratio v_1/v_2?
   A. M_1/M_2
   B. M_2/M_1
   C. $(M_1/M_2)^{1/2}$

D. $(M_2/M_1)^{1/2}$

6. A sphere of mass m at the end of a string of length ℓ is released when the string is horizontal. The sphere then collides with a block of mass $4m$ at rest on a smooth table. If the collision in perfectly elastic, find in terms of m, ℓ and g or some of them,

   (a) velocity of the sphere and

   (b) the velocity of the block after collision.



Solution:

(a)

$$-\frac{3}{5}\sqrt{2gl}$$

(b)

$$\frac{2}{5}\sqrt{2gl}$$

7. A 2150 kg car moving east at 10.0 m/s collides with a 3250 kg car moving east. The cars stick together and move as a unit after the collision at a velocity of 5.22 m/s.

   (a) Find the velocity of 3250 kg car before the collision.

   (b) What is the decrease in kinetic energy during the collision?

Solution:

(a) 2.1 m/s

(b) 4.1×10^4 J

8. An unstable nucleus with a mass of 17.0×10^{-27} kg initially at rest disintegrate into 3 particles. One of the particle of mass 5.0×10^{-27} kg, moves along the positive $y-$axis with a speed of 6.0×10^6 m/s. Another particle, of mass 8.4×10^{-27} kg, moves along the positive $x-$axis with a speed of 4.0×10^6 m/s. Determine the third particle's speed and direction of motion.

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Solution: $v_{3f} = 1.3 \times 10^7$ m/s; 41° below the negative axis.

9. A bullet of mass $m = 12\ g$ strikes a stationary wooden block of mass $M = 5.2\ kg$ standing on a frictionless surface. The block, with the bullet embedded in it, acquires a velocity of $v = 1.7\ m/s$.
   (a) What was the velocity of the bullet before it struck the block?
   (b) What fraction of the bullet's initial kinetic energy is lost (i.e., dissipated) due to the collision with the block?

Solution:
(a) $738.4\ m/s$
(b) $f = K_i/(K_i - K_f) = 0.9977$

10. Two objects slide over a frictionless horizontal surface. The first object, mass $m_1 = 5\ kg$, is propelled with speed $v_{i1} = 4.5\ m/s$ toward the second object, mass $m_2 = 2.5\ kg$, which is initially at rest. After the collision, both objects have velocities which are directed $\theta = 30^\circ$ on either side of the original line of motion of the first object.
   (a) What are the final speeds of the two objects?
   (b) Is the collision elastic or inelastic?

Solution:
(a) $v_{f1} = \dfrac{v_{i1}}{2.\cos\theta} = 2.5981\ m/s$; $v_{f2} = \dfrac{m_1}{m_2} v_{f1} = 5.1962\ m/s$
(b) $K_i = K_f \Rightarrow$ the collision is elastic.

11. A softball of mass $m = 0.35\ kg$ is pitched at a speed of $u = 12\ m/s$. The batter hits the ball directly back to the pitcher at a speed of $v = 21\ m/s$. The bat acts on the ball for $t = 0.01\ s$.
   (a) What impulse is imparted by the bat to the ball?
   (b) What average force is exerted by the bat on the ball?

Solution:
(a) $11.55\ Ns$
(b) $1155.0\ N$

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12. What is the change of energy while operating a light bulb?
    A. chemical $\rightarrow$ heat
    B. electrical energy $\rightarrow$ heat & light
    C. heat & light $\rightarrow$ electrical energy
    D. electrical energy $\rightarrow$ kinetic energy

13. What are the main energy changes in a battery-operated remote control car?
    A. chemical $\rightarrow$ electrical $\rightarrow$ kinetic
    B. chemical $\rightarrow$ wave $\rightarrow$ kinetic
    C. wave $\rightarrow$ electrical $\rightarrow$ kinetic
    D. electrical $\rightarrow$ chemical $\rightarrow$ wave

14. A battery supplies 500 J of electrical energy to a motor such that an object moves horizontally for 40 m under the influence of a 6 N pulling force. What is the efficiency of the motor?
    A. 21%
    B. 24%
    C. 48%
    D. 75%

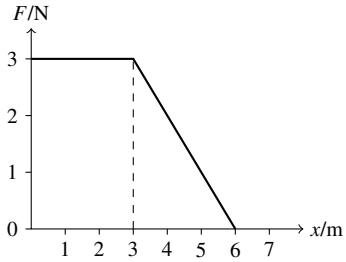
15. A particle is displaced from position $\overrightarrow{r}_1 = (3\overrightarrow{i} + 2\overrightarrow{j} - 6\overrightarrow{k})$ to another position $\overrightarrow{r}_2 = (14\overrightarrow{i} + 13\overrightarrow{j} + 9\overrightarrow{k})$ under the impact of a force $5\overrightarrow{i}$ Newton. The work done will be given by
    A. 55 J
    B. 65 J
    C. 45 J
    D. 75 J

Solution:

$$\begin{aligned}\text{Work} &= \overrightarrow{\text{Force}} \times \overrightarrow{\text{Displacement}} \\ &= 5\overrightarrow{i} \times (11\overrightarrow{i} + 11\overrightarrow{j} + 15\overrightarrow{k}) \\ &= 55 \text{ J}.\end{aligned}$$

16. A force F acting on an object varies with distance x as shown in the below figure. The force is in N and x in m. The work done by the force in moving the object from $x = 0$ to $x = 6$ m is

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A. 18 J

B. 13.5*J*

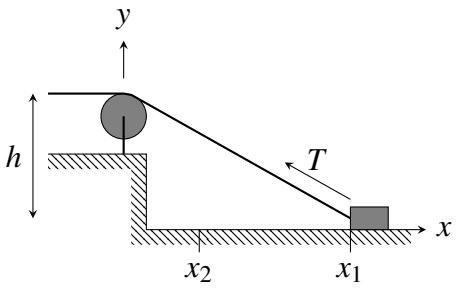
C. 9 J

D. 4.5 J

Solution: Work $= \int \vec{F}.\vec{dx}$ = The area enclosed between $(F-x)$ curve = $\frac{1}{2}(6+3).3 = 13.5$ J

Choose the correct answer: 13.5 J

17. Figure shows a cord attached to a cart that can slide along a frictionless horizontal rail aligned along an x axis. The left end of the cord is pulled over a pulley, of negligible mass and friction and at cord height $h = 1.20$ m, so the cart slides from $x_1 = 3.00$ m to $x_2 = 1.00$ m. During the move, the tension in the cord is a constant 25.0 N. What is the change in the kinetic energy of the cart during the move?



A. 20.5 J

B. 33.7 J

C. 41.7 **J**

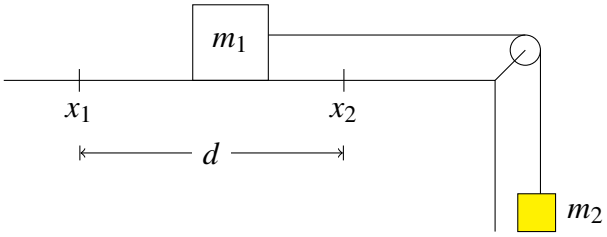
D. 51.4 J

Solution:

$$\Delta K = F.d = F\left(\sqrt{x_1^2+h^2} - \sqrt{x_2^2+h^2}\right) = 25.0\left(\sqrt{3^2+1.2^2} - \sqrt{1^2+1.2^2}\right) = 41.7\text{J}$$

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18. In the experimental setup below, two masses, m_1 and m_2, are connected by a massless string over a massless pulley. Mass m_1 slides on a frictionless surface. The values of the two masses can be measured, as well as the distance d and the speed of mass m_1 as it passes x_1 and again ar x_2. The experiment can be used to do which of the following?



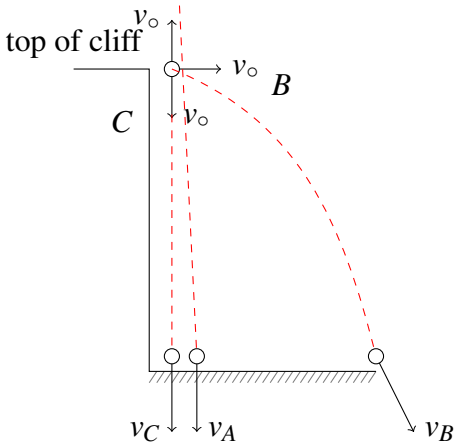
1. Demonstrate momentum conservation
2. Demonstrate energy conservation
3. Measure the value of the acceleration due to gravity

A. 1 only
B. 3 only
C. 1 and 2
D. 2 and 3

Solution: Answer is D

19. Suppose a ball is thrown off a cliff with the same initial speed, v_0, in the following three manners (see Figure):

A. The ball is thrown upward.
B. The ball is thrown horizontally.
C. The ball is thrown downward.



Here, v_A, v_B and v_C denote the speeds of the ball at the moment when it reaches the ground in cases A, B and C, respectively. Choose the best relation between v_A, v_B and v_C, assuming that air resistance is negligible.

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A. $v_A > v_B > v_C$
B. $v_B > v_A > v_C$
C. $v_A = v_C > v_B$
D. $v_A = v_B = v_C$

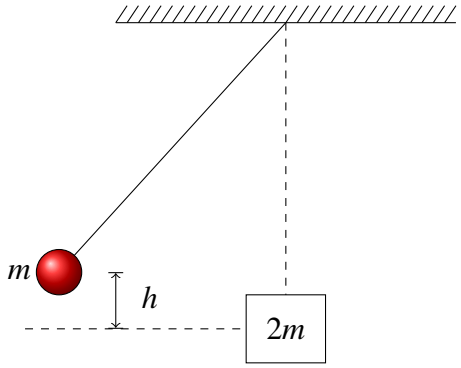
Solution: Answer is D.

- In all of the three cases, the ball is initially located at the same height, h, and moving at the same speed, v_0. Using the law of conservation of energy, we get the velocity of the ball, v, at the instant it reaches the ground for each of the cases as follows:

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_0^2 + mgh \text{ with } v = \sqrt{v_0^2 + 2gh}$$

- Therefore, we obtain $v_A = v_B = v_C$

20. As shown below, a ball of mass m, suspended on the end of a wire, is released from height h and collides elastically, when it is at its lowest point, with a block of mass $2m$ at rest on a frictionless surface. After the collision, the ball rises to a final height equal to



A. $1/9h$
B. $1/8h$
C. $1/3h$
D. $2/3h$

Solution:

- Conservation of momentum and kinetic energy of the system (with $m_A = m,\ m_B = 2m,\ v_B = 0$)

$$m_A v_A + m_B v_B = m_A v'_A + m_B v'_B$$

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$$\frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2 = \frac{1}{2}m_A v_A' + \frac{1}{2}m_B {v_B'}^2$$

- From the two equation above, we have:

$$v_B' = 2v_A'$$

$$v_A = -3v_A'$$

- Conservation of energy at the moment of the collision:

$$U = T \rightarrow m_A gh = \frac{1}{2}m_A v_A^2 \rightarrow v_a = \sqrt{2gh}$$

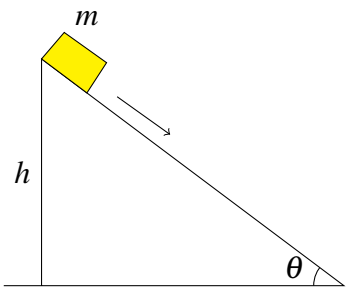
- After the collision:

$$U' = T' \rightarrow m_A gh' = \frac{1}{2}m_A {v_A'}^2 \rightarrow v_A' = \sqrt{2gh'}$$

- Combine the equations, we find that:

$$h' = \frac{1}{9}h$$

21. A block of mass m sliding down an incline at constant speed is initially at a height h above the ground, as shown in the figure below. The coefficient of kinetic friction between the mass and the incline is μ. If the mass continues to slide down the incline at a constant speed, how much energy is dissipated by friction by the time the mass reaches the bottom of the incline?



A. mgh/μ

B. mgh

C. $\mu mgh/\sin\theta$

D. $mgh\sin\theta$

Solution: Answer is B

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22. A block of mass $m = 3\ kg$ starts at rest at a height of $h = 43\ cm$ on a plane that has an angle of inclination of $\theta = 35^\circ$ with respect to the horizontal. The block slides down the plane, and, upon reaching the bottom, then slides along a horizontal surface. The coefficient of kinetic friction of the block on both surfaces is $\mu = 0.25$. Using the *work-kinetic energy theorem*, determine how far does the block slide along the horizontal surface before coming to rest?

Solution: 1.10 m

23. A 25 (kg) child on a 2.0 (m) long swing is released from rest when the swing supports make an angle of 30° with the vertical.
    (a) What is the maximum potential energy associated with the child?
    (b) Disregarding friction, find the child's speed at the lowest position.
    (c) What is the child's total mechanical energy?
    (d) If the speed of the child at the lowest position is 2.00 (m/s), what is the change in mechanical energy due to friction?

Solution:

(a) 65.7 J

(b) 2.3 m/s

(c) 65.7 J

(c) -16 J

24. A pirate drags a 50 kg treasure chest over the rough surface of a dock by exerting a constant force of 95 N acting at an angle of 15° above the horizontal. The chest moves 6 m in a straight line, and the coefficient of kinetic friction between the chest and the dock is 0.15. How much work does the pirate perform? How much energy is dissipated as heat via friction? What is the final velocity of the chest?

Solution:

- Work done by the pirate: 550.6 J
- Energy dissipated as heat due to friction: -419.3 J
- Final velocity of the chest: 2.29 m/s

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25. Starting from rest, a 10.0 (kg) suitcase slides 3.00 (m) down a frictionless ramp inclined at 30.0° from the floor. The suitcase then slides an additional 5.00 (m) along the floor before coming to a stop. Determine the following:

    (a) the speed of the suitcase at the bottom of the ramp

    (b) the coefficient of kinetic friction between the suitcase and the floor.

    (c) the change in mechanical energy due to friction.

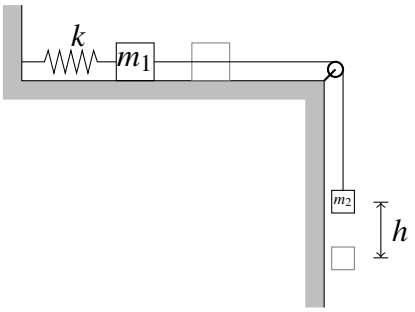
Solution:

(a) 5.42 m/s

(b) 0.3

(c) -147 J

26. Two blocks are connected by a light string that passes over a frictionless pulley, as shown in the below figure. The block of mass m_1 lies on a horizontal surface and is connected to a spring of force constant k. The system is released from rest when the spring is unstretched. If the hanging block of mass m_2 falls a distance h before coming to rest, calculate the coefficient of kinetic friction between the block of mass m_1 and the surface.



Solution:

$$\mu = \frac{m_2 g - kh/2}{m_1 g}$$

27. Which word equation correctly defines power?

    A. force × distance moved in the direction of the force

    B. force × velocity

    C. work done : time taken

    D. work done × time taken

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28. A body of mass m is moved a distance s at a steady speed v against a constant force F. What is the power required to keep up this motion?

    A. mv
    B. $mv^2/2$
    C. Fv
    D. Fs

Solution: Answer is C

29. A car engine provides 24 kW of useful power in order to travel at a constant speed. If the driving force on the car is 600 N, what is the speed of the car?

    A. 2.5 m s^{-1}
    B. 4.0 m s^{-1}
    C. 25 m s^{-1}
    D. 40 m s^{-1}

30. When the hook of a crane is lifted at a steady velocity, a force of 1000 N is used. A load of mass 1000 kg is now lifted by the crane at a velocuty of 0.5 m s^{-1}. [Take g as 10 m s^{-2}]
    How much power is used to lift the hook and the load?

    A. 5.0 kW
    B. 5.5 kW
    C. 20 kW
    D. 22 kW

31. A body of mass m is moving up a frictionless track inclined at an angle θ to the horizotal. What is the power required to give the body a forward acceleration a when its velocityis v ?

    A. $mavg\sin\theta$
    B. $mav\sin\theta + mgv$
    C. $mav + mgv\sin\theta$
    D. $(mav + mgv)\sin\theta$

Solution: Answer is C

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